Momentum

antocciphysics | TEACHER KEY | CPA

Learning target: Calculate and compare signed momentum.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 15; 3 course days later.*
Need orbital geometry/positions and phase or time reference. A period alone does not show simultaneous nearby positions; uncertainties matter.

02. Explain a worked example (worked)
Both left; velocity 2 m/s, momentum 6 kg·m/s.

03. Look and explain (model)
Rightward cart positive; leftward cart negative.

04. Build the solution (practice)
+6 and -6; total 0 kg·m/s. Both are moving.

05. Use the idea (reason)
Both double momentum magnitude if direction unchanged.

06. Choose a method (mixed) *Review: Unit 2, Day 6; 12 course days later.*
g=6 N/kg; force 48 N toward the source.

07. Exit check (exit)
8 kg·m/s left.

08. STRETCH: Apply it another way (stretch)
Example 1 kg at +6 m/s and 3 kg at -2 m/s; momenta +6 and -6.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 15; 3 course days later.*
Overlap days 15-17. Timing is compatible, but spatial geometry, miss distance and uncertainties are still needed.

02. Explain a worked example (worked)
Both left; velocity 2 m/s, momentum 6 kg·m/s.

03. Represent and interpret (model)
Rightward cart positive; leftward cart negative.

04. Solve and justify (practice)
+3 and -3 kg·m/s; total zero. Equal opposite momentum arrows cancel, although both carts move.

05. Analyze the reasoning (reason)
Both double momentum magnitude if direction unchanged.

06. Choose a method (mixed) *Review: Unit 2, Day 6; 12 course days later.*
Field 4 N/kg; force 3.2 N toward the source.

07. Exit check (exit)
v=-4 m/s; new p=+12 kg·m/s.

08. STRETCH: Apply it another way (stretch)
4-10+3 v=0 gives v=+2 m/s. After reversing first cart, total -4-10+6=-8 kg·m/s.